

Unit 1, Lesson 4: Evaluating Numeric Expressions – Part 1

Benchmark

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

Learning Target

☐ I can evaluate numeric expressions with up to four operations.

1.4.1 Warm-Up: Notice and Wonder

In 1672, the Castillo de San Marcos was built by the Spanish in St. Augustine, Florida to defend Florida and the Atlantic trade route.



1. What do you notice about the design of the Castillo de San Marcos?
2. What do you wonder?

1.4.2 Refresher: Order of Operations

1. Work with a partner to number each operation so that the order of operations is in the correct order.



Perform operations inside grouping symbols.



Perform multiplication or division in order from left to right.
(→)



Evaluate expressions with exponents.



Perform addition or subtraction in order from left to right.
(→)

1.4.3 Guided Practice: Using the Order of Operations to Evaluate Numeric Expressions

1. For each table, an expression is evaluated, but the steps are mixed up. Order the expressions by writing the numbered step next to each expression shown. Explain the steps in the column on the right. The first table has been started.

a. Complete the table.

Step	Expression	Explanation
	0.45	
2	$-6.75 \div 15 \cdot (-1)$	Evaluate the exponent
1	$-6.75 \div 15 \cdot (-1)^5$	Original expression
	$-0.45 \cdot (-1)$	

b. Complete the table.

Step	Expression	Explanation
	$ 5.76 - 17.75$	
1	$ (9.2 - 11.6)^2 - 17.75$	Original expression
	-11.99	
	$5.76 - 17.75$	
	$ (-2.4)^2 - 17.75$	

2. Evaluate each expression using the order of operations.

a. $(-5 + 4.5)^3 - |-8.7|$

b. $\left(\frac{41-25}{-2}\right)^2 \times (-9)$

c. $-7.1 + 2|1.46 - 9.2|$

1.4.4 Collaboration: Using Order of Operations

1. Amiyah and David simplified the same expression using an order of operations and arrived at two different solutions. Their work is shown.

Amiyah	David
$-10 - 7(25^2) - -5\frac{3}{4} $ $-10 - 7(25^2) - 5.75$ $-10 - 7(6.25) - 5.75$ $-17(6.25) - 5.75$ $-106.25 - 5.75$ -112	$-10 - 7(25^2) - -5\frac{3}{4} $ $-10 - 7(6.25) - -5.75 $ $-10 - 43.75 - -5.75 $ $-10 - 43.75 + 5.75$ -48

- a. Amiyah and David both made common mistakes. Circle the mistakes in the work.
- b. Write a note to Amiyah and David explaining the mistake(s) they made.

Amiyah	David

- c. Evaluate the expression correctly.

1.4.5 Activity: Evaluating Expressions

1. Decide with your partner who will be Partner A and who will be Partner B.
 - a. Evaluate the problems in your column.

Partner A	Partner B
$12 - -6.5 + 3^2 $	$9 - 5^2 - 8.4 $
$\left(\frac{3-75}{15}\right)^2 \times 8$	$\left(\frac{45-75}{5}\right)^2 \times -\frac{1}{2}$
$(-32) \div (-2)^4 \times (-3.6 - 1.2)$	$(-0.74 \times 5) + 48 \div (-2)^2$

$ (30.5 - 5^2) \div 2 $	$ (23.5 - 4^3) \div 10 $
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- b. Calculate the sum of the answers in each column from Part A.

Sum of Partner A's Answers	Sum of Partner B's Answers

- c. Both columns should have the same sum. If that is not the case, work with your partner to find and correct any errors in your work.

1.4.6 Your Turn!

1. Evaluate each expression. Show the steps in your work.

a. $-4.5 - |(3 \times -1.6)^2|$

b. $\frac{(-7^6 \times 0.2)^4}{-2}$

c. $(-18.6) \div 6 \cdot (-2.3 + 5.4)^2$

d. $[(-3)^2 \times (3.7 - 5.8)]^3$

1.4.7 Wrap-Up: Cool Down

1. For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	
When you add two positive decimals together, the value is always larger.	
To add exponents, their bases must be the same.	
The product of two fractions is always a larger number.	
The sum of a positive and a negative number is always negative.	